

The Impact of Stadium Attendance on Premier League Home Advantage During and Post-COVID-19

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<https://github.com/UVA-STAT3080/project-sp26-cody-c.git>

Introduction

Home advantage, the tendency for teams to perform better at their own stadium, is one of the most consistent patterns in professional soccer. Fan support has long been proposed as a key driver of this effect, with crowds thought to boost home player confidence, intimidate opponents, and subtly influence referee decisions. The COVID-19 pandemic provided a rare natural experiment: Premier League matches in the 2019–20 and 2020–21 seasons were played with little or no spectators, temporarily stripping away crowd support.

In Part 1, we explored home win percentages across three periods: Pre-COVID (2016–17 through 2018–19), COVID (2019–20 and 2020–21), and Post-COVID (2021–22 through 2023–24). The exploratory analysis revealed that mean home win percentage declined from 47.2% in the pre-COVID era to 41.3% during COVID, before partially recovering to 43.7% post-COVID. The boxplots and trend line both showed a clear dip during the COVID period with incomplete recovery afterward.

The focused research question for Part 2 is: **Is there a statistically significant difference in the distribution of home win percentage between the COVID-affected seasons and the post-COVID seasons in the Premier League?**

Methods

Procedure: We apply a **Wilcoxon rank-sum test** (also known as the Mann–Whitney U test) to compare the distribution of home win percentage between the COVID group ($n = 40$

team-seasons) and the Post-COVID group ($n = 60$ team-seasons). This is the nonparametric alternative to the Welch two-sample t -test. The hypotheses are:

H_0 : The distributions of home win % are equal across COVID and Post-COVID periods

H_a : The distributions of home win % differ between COVID and Post-COVID periods

Justification of appropriateness: The Wilcoxon rank-sum test requires three assumptions: (1) the two samples are independent of each other, (2) observations within each group are independent, and (3) the data are at least ordinal (continuous data satisfies this). We verify each assumption for both groups below.

Independence between groups: The COVID group (2019–20 and 2020–21 seasons) and the Post-COVID group (2021–22 through 2023–24 seasons) represent entirely separate, non-overlapping sets of seasons. No team-season observation appears in both groups, so the two samples are independent of each other.

Independence within each group: Each observation in both the COVID group and the Post-COVID group is a unique team-season combination. Within a given season, teams do share opponents, but across seasons each team-season record is recorded independently. This is the standard unit of analysis in sports performance research and the same unit used throughout Part 1. This holds for both the COVID and Post-COVID groups.

Continuous data: Home win percentage is a continuous, ratio-scale variable for both groups, calculated as total home wins divided by home games played, multiplied by 100. This satisfies the ordinal data requirement for both populations.

Normality (checked to justify nonparametric choice): The Wilcoxon rank-sum test does not require normality, but we verify via Q-Q plots below that normality cannot be assumed for at least one group, confirming the nonparametric approach is necessary rather than just optional.

The COVID group Q-Q plot shows a clear outlier in the upper tail that departs substantially from the reference line, indicating a violation of normality for that population. The Post-COVID group follows the reference line more closely through the middle of the distribution but shows some departure at both tails. Because normality cannot be reasonably assumed for the COVID group, the Wilcoxon rank-sum test is the more appropriate procedure for both groups. All three assumptions of the Wilcoxon rank-sum test are satisfied for each population.

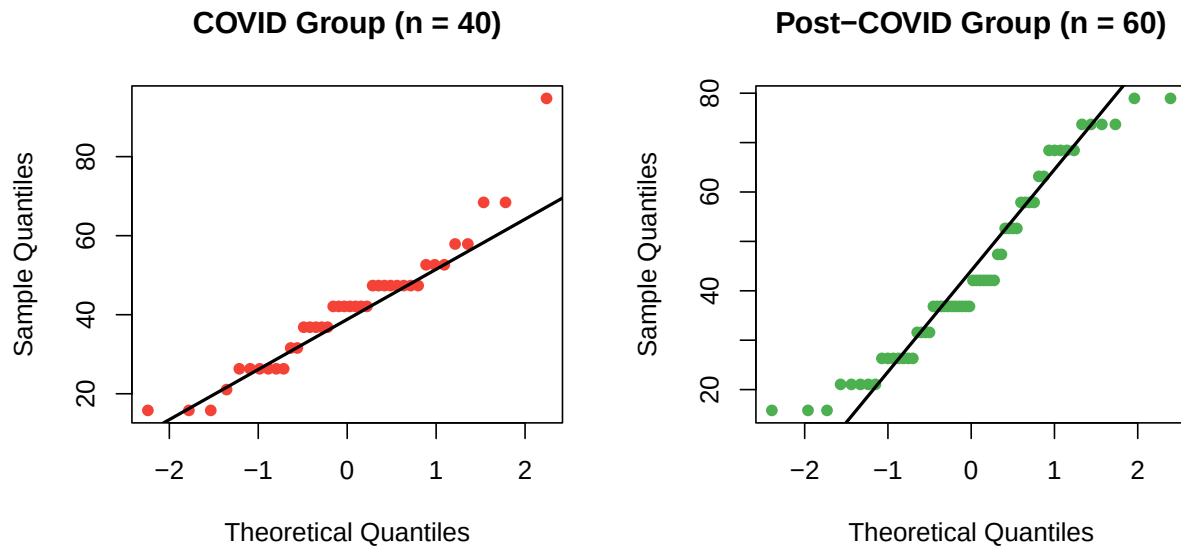


Figure 1: Normal Q-Q plots of home win percentage for the COVID (left) and Post-COVID (right) groups. The COVID group shows a notable departure from normality in the upper tail, violating the normality assumption required for a t-test.

Results

```
# Wilcoxon rank-sum test: COVID vs Post-COVID home win percentage
test_result <- wilcox.test(Home_Win_Pct ~ Group,
                           data      = df_test,
                           alternative = "two.sided",
                           exact     = FALSE)

test_result
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: Home_Win_Pct by Group
## W = 1150, p-value = 0.7261
## alternative hypothesis: true location shift is not equal to 0
```

The Wilcoxon rank-sum test produced a test statistic of $W = 1150$ and a p-value of 0.726. The test statistic W represents the sum of ranks from the COVID group; values far from the

expected value under H_0 indicate a distributional shift between the two groups. The p-value represents the probability of observing a test statistic at least as extreme as $W = 1150$ if the two distributions were truly equal.

At the conventional significance level of $\alpha = 0.05$, **the test is not statistically significant** ($p = 0.726 > 0.05$). We fail to reject H_0 and cannot conclude that the distribution of home win percentages differs between the COVID and Post-COVID periods.

Conclusions

Despite the exploratory analysis showing a visible gap in mean home win percentage between the COVID period (41.3%) and the post-COVID period (43.7%), the Wilcoxon rank-sum test indicates this difference is not statistically significant ($p = 0.726$). In practical terms, we do not have sufficient statistical evidence to conclude that the distribution of home win percentage shifted after fans returned to stadiums, relative to the COVID baseline. The return of fans to Premier League stadiums cannot be said to have produced a statistically detectable recovery in home advantage based on this data.

Limitations: Several factors constrain the strength of these conclusions. First, the COVID group spans only two seasons ($n = 40$ team-seasons), limiting statistical power and increasing the chance of a Type II error. Second, team composition changes substantially across seasons through promotion, relegation, and transfers, so individual team-seasons are not identically distributed within each group. Third, the 2019–20 season was only partially played without fans, introducing heterogeneity into the COVID group. Finally, we cannot isolate fan absence as the sole cause of any difference; changes in team quality, tactical adjustments, and lingering pandemic effects all contribute.

Generalizability: To the extent that the Premier League represents elite professional soccer broadly, the finding that home advantage did not significantly recover in the post-COVID seasons may be relevant to other top European leagues. However, caution is warranted: leagues differed in how quickly and fully fans were permitted to return, and structural differences across leagues may produce different patterns.

References

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Wunderlich, F., Weigelt, M., Rein, R., & Memmert, D. (2021). How does spectator presence affect football? Home advantage remains in European top-class football matches played without spectators during the COVID-19 pandemic. *PLOS ONE*, *16*(3), e0248590. <https://doi.org/10.1371/journal.pone.0289899>

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